FLASHING INSTRUCTION FOR NRF52840

There are 2 ways to do this either using J link flash software or nrf connect vscode.

1. USING J-FLASH: (J LINK ADAPTER):

1. download segger J-Link from https://www.segger.com/downloads/jlink/
2. Connect the nrf52840 through the J-link adapter to your PC .

J-Link Pin	Signal Name	nRF52840 Pin
1 (VTref)	VCC (Target Voltage)	VDD (3.3V)
4 (GND)	Ground	GND
7 (SWDIO)	SWD Data	SWDIO
9 (SWCLK)	SWD Clock	SWDCL
15 (nRESET) [Optional]	Reset	P0.18 (RESET)

3. Open J-Link Flash and create a new project
4. A prompt will open asking for target device , select nrf52480 as target device and leave the other setting as it is.
5. now drag the merged.hex file from the (flashing_guide->bin) folder to the box asking for hex file or you can also select it from file->Open data file
6. now navigate to target-> and press Connect
7. now navigate to target->manual programming-> program and verify
8. if it shows target programmed and verify successfully ,you can run the application by target->manual programming-> Start Application

Note: if the flashing fails , try to erase the chip by navigating and selecting target->manual programming-> Erase chip and then reprogramming it again.

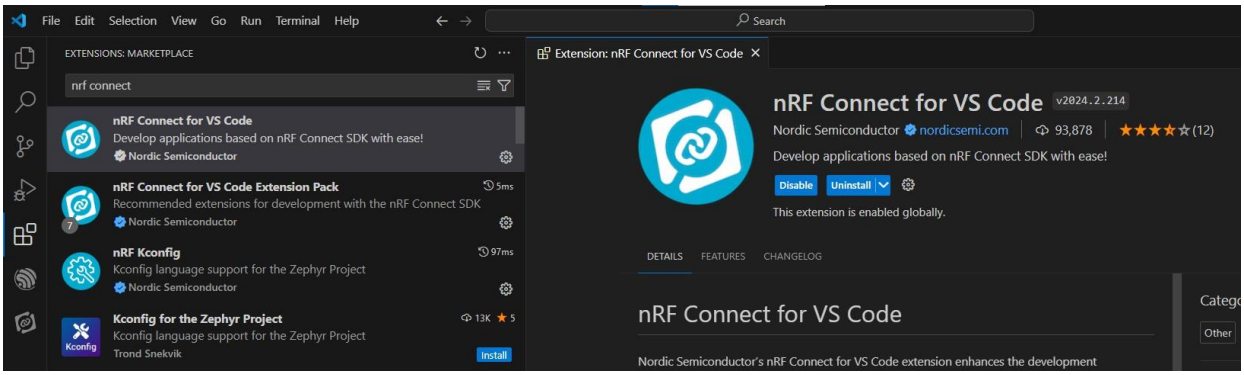
2. USING NRF CONNECT SDK (VSCODE):

1 - Setup NRF connect SDK on VS Code

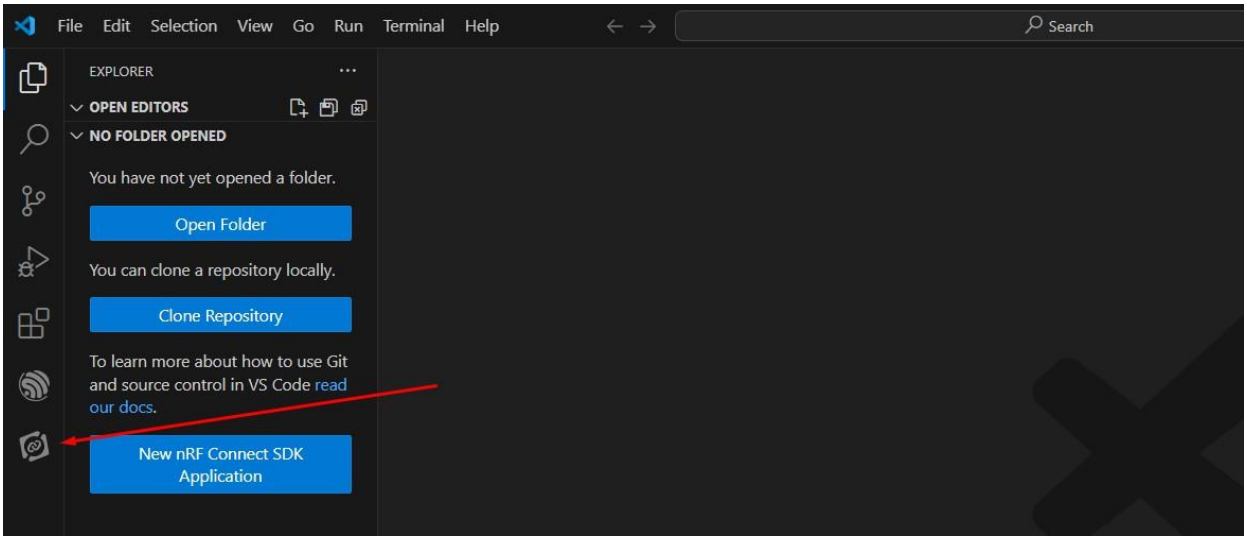
A. VS Code Installation
   1. Download the Visual Studio Code installer for Windows.
   2. Once it is downloaded, run the installer (VSCodeUserSetup-{version}.exe). This will only take a minute.
   3. Open VS Code.

B. Add NRF Connect SDK extension in VS Code.

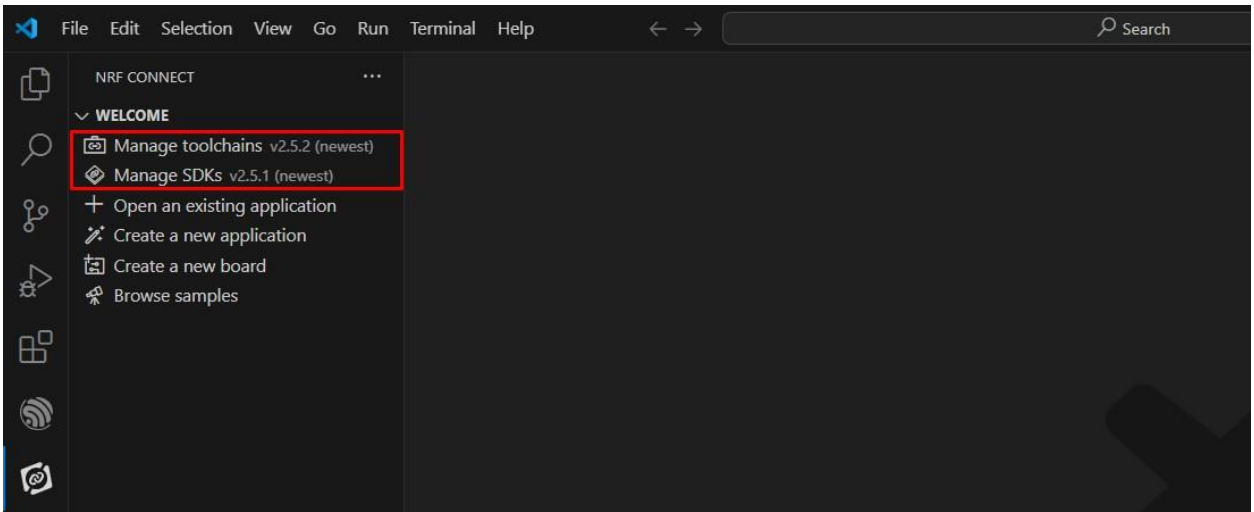
1. Add both the nrf connect sdk extensions.



2. After installing the extensions nrf connect extension icon will appear on the left bar.

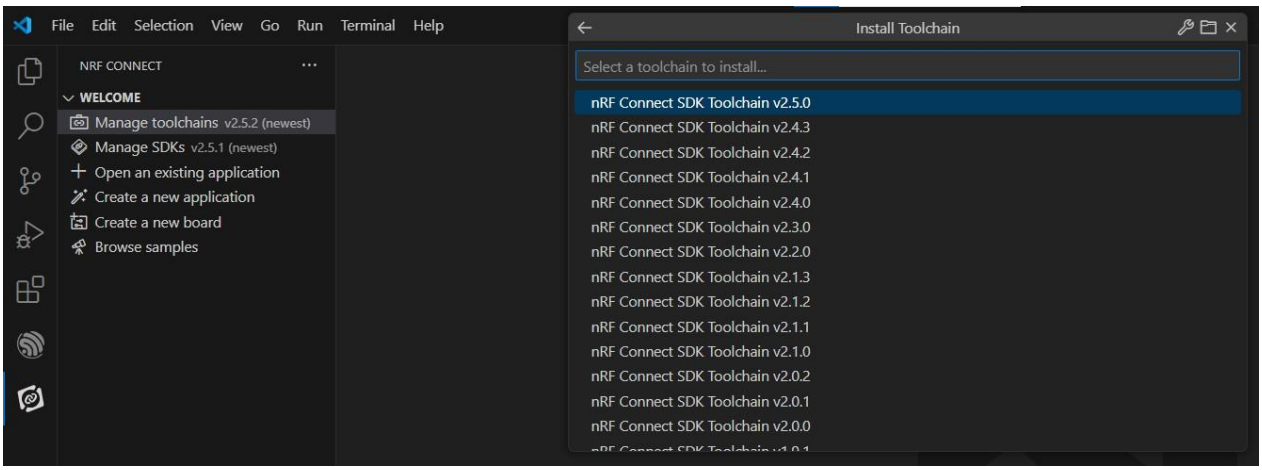


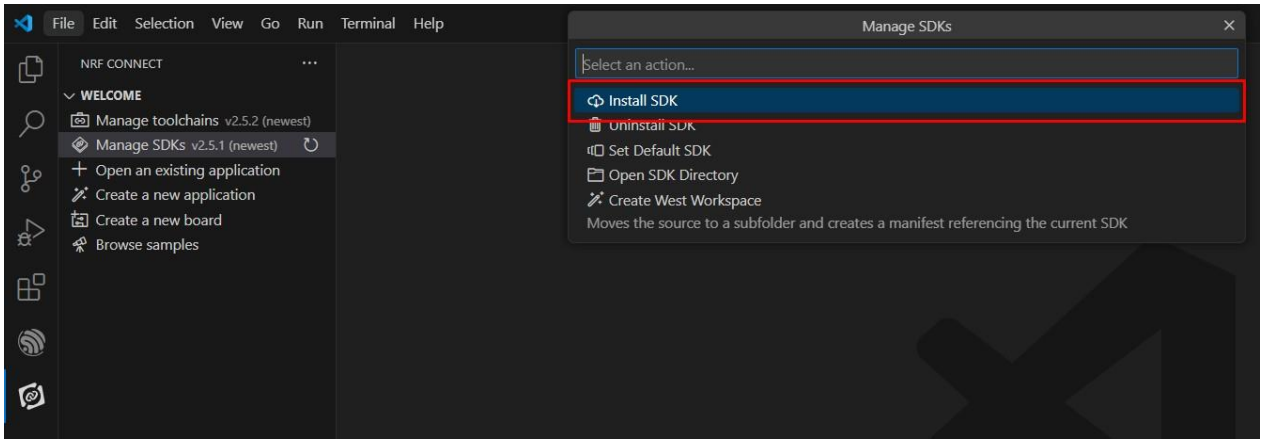
Install Toolchain- lick on Manage toolchain, then install the nRF Connect SDK Toolchain v2.7.0.

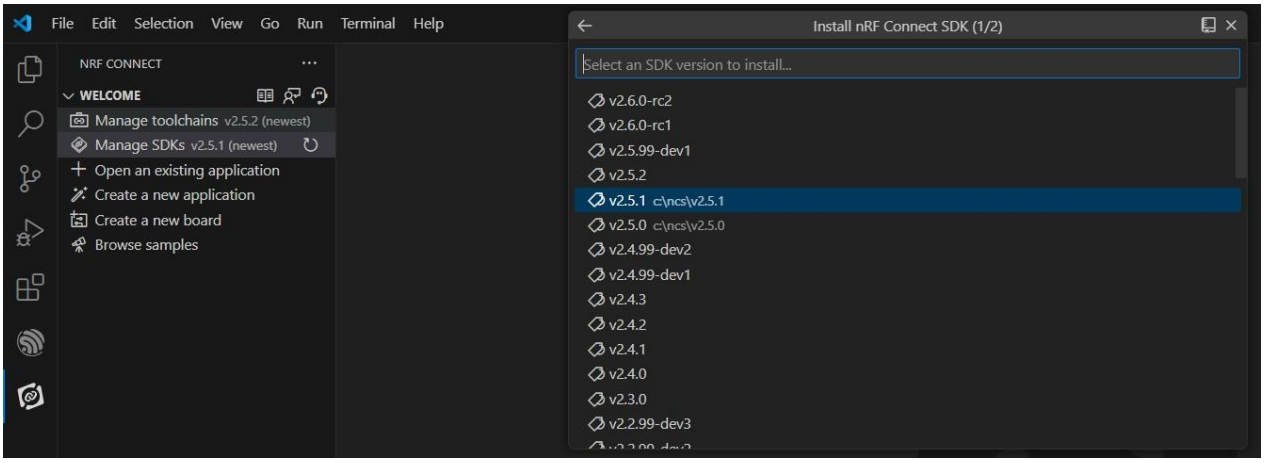


-Install SDK

Then do the same for Manage SDKs, install v2.7.0







2 - Install J-Link tools

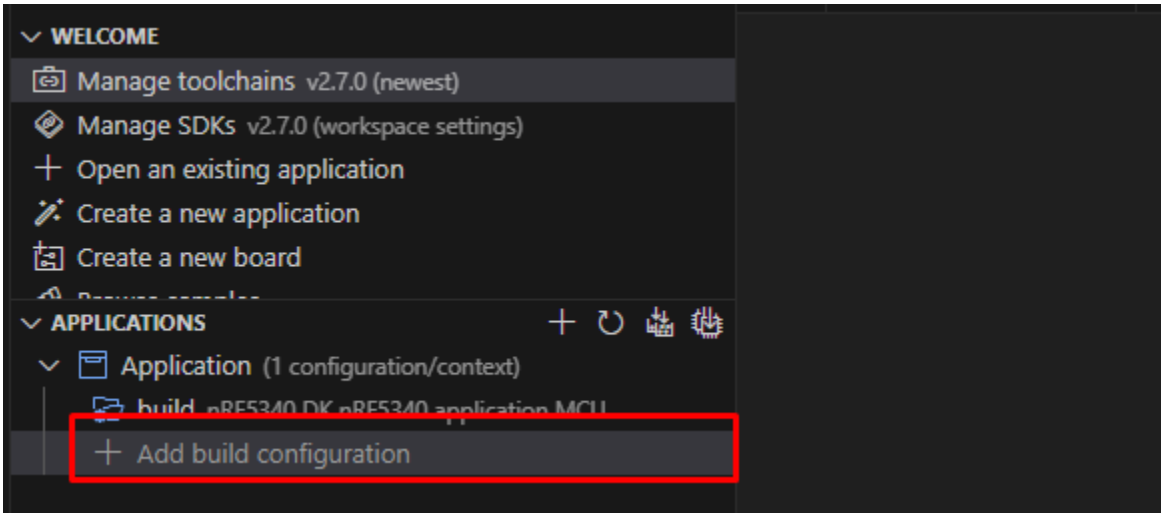
Download J-Link tools from here We used v7.94e. Download the latest from this link for the tools here

- Follow the installation guide and install

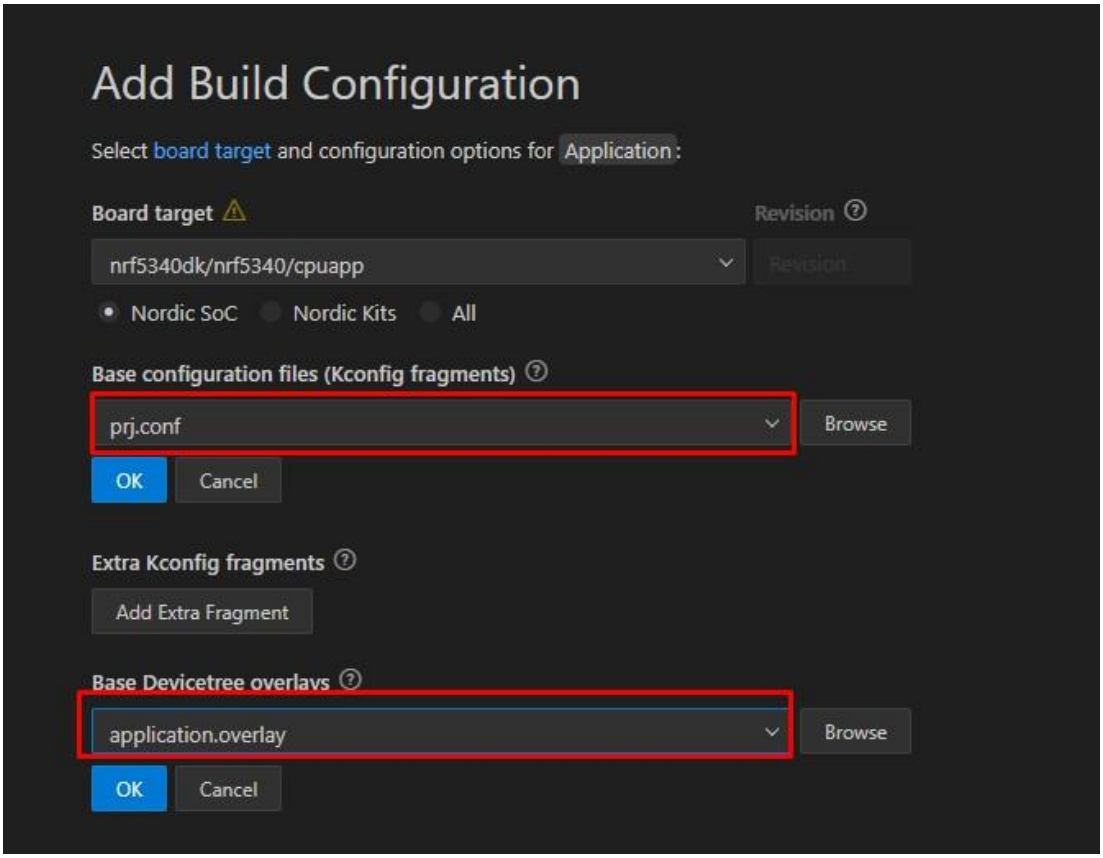


3 - Flashing the Device

1. Go to NRF Connect Extention Tab. Select “Add Build Configuration”.



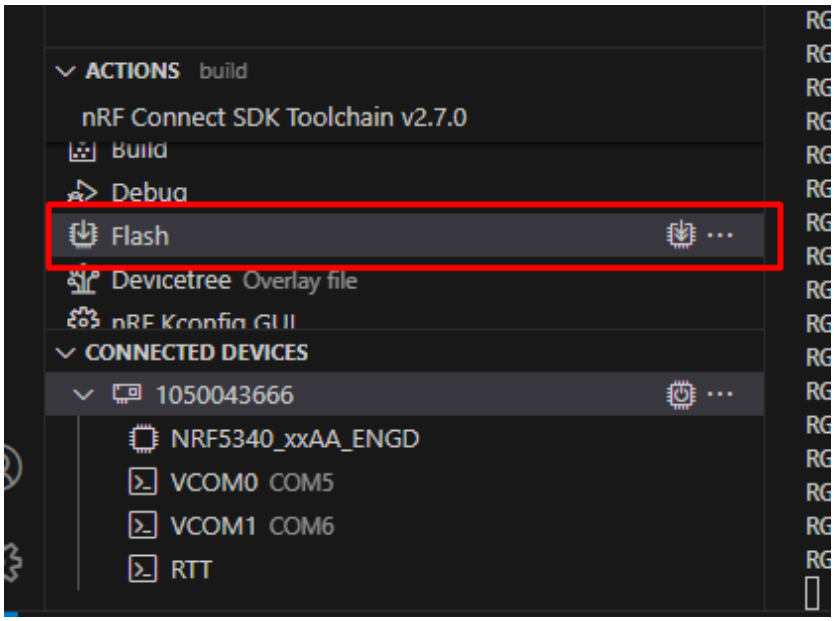
2. Select “prj.conf” as Base configuration files and “application.overlay” as Base Devicetree overlay.



3. Click Build Configuration.

4. After building , replace the merged.hex file with the given one.

5. Once build, the device can be flashed from the Actions section.



MONITORING LOGS FOR NRF52840

USING UART TO USB BRIDGE/ADAPTER (CP210X):

1. install the CP210x driver using this link
2. now plugin the adapter to your pc
3. connect the txd pin of the adapter to the TRACE DATA 2 pin (p0.11)
4. connect the rxd pin of the adapter to the TRACE DATA 1 pin (p0.12)
5. now go to device manager->ports and see if the device is detected , if its already there then its okay if not then update driver and select the CP210x.zip file downloaded previously
6. you will be able to monitor logs using any serial monitor and selecting the COM port at which Silicon Labs Cp210x is present and setting the baud rate to 115200 .
7. Logs will only appear after the merged.hex file is flashed and the NRF52840 is powered on.

On successfull setup following logs will be printed :(***Note: if the logs are not showing press reset the nrf and do it while monitoring***)

```
I2C1 initialized successfully!

Scanning I2C1 bus...

Sensor detected at 0x77 Bus:I2C1

I2C1 scan complete.

TLV initialization successfull!

TLV setup started

I2C readout failed in TLV493_ReadOut with i2c error code -5!

TLV setup failed!

*** Booting nRF Connect SDK v2.9.0-7787b2649840 ***

*** Using Zephyr OS v3.7.99-1f8f3dc29142 ***

I2C1 initialized successfully!

Scanning I2C1 bus...

Sensor detected at 0x77 Bus:I2C1

I2C1 scan complete.

TLV initialization successfull!

TLV setup started

I2c readout successfull in TLV493_ReadOut!
```

```
buff value in TLV493 Readout:   00800000800000800080

setting default config

I2c writeout successfull in Tlv493d_WriteOut!

buff value in TLV493 write_read :       29aa164b

Tlv493d_SetDefaultConfiguration successful in setup!

I2c writeout successfull in Tlv493d_WriteOut!

buff value in TLV493 write_read :       29aa164b

TLV setup successfull!

updating data

I2c readout successfull in TLV493_ReadOut!

buff value in TLV493 Readout:   1300461f000001

X_value = 48

Y_value = 0

Z_value = 96

Temp_value = 1
```

if the sensor fails to detect:

```
I2C1 initialized successfully!
Scanning I2C1 bus...
Sensor detected at 0x77 Bus:I2C1
I2C1 scan complete.
TLV initialization successfull!
TLV setup started
I2C readout failed in TLV493_ReadOut with i2c error code -5!
TLV setup failed!
```